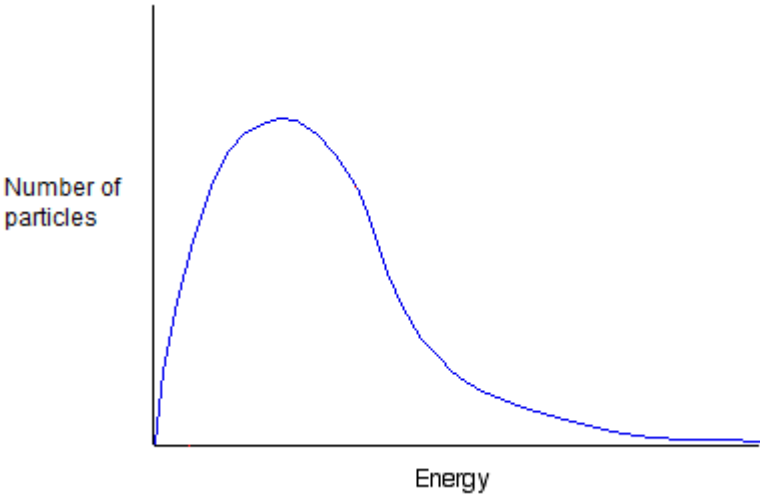


Question				Marking details		Marks available																	
						AO1	AO2	AO3	Total	Maths	Prac												
12.	(a)	(i)		<table border="1"><tr><td>[I⁻]</td><td>Rate</td></tr><tr><td>0.1</td><td>18</td></tr><tr><td>0.2</td><td>50</td></tr><tr><td>0.3</td><td>56</td></tr><tr><td>0.4</td><td>83</td></tr><tr><td>0.5</td><td>100</td></tr></table>		[I ⁻]	Rate	0.1	18	0.2	50	0.3	56	0.4	83	0.5	100		1		1	1	1
				[I ⁻]	Rate																		
				0.1	18																		
				0.2	50																		
				0.3	56																		
				0.4	83																		
				0.5	100																		
		(ii)		axes – labelled and minimum half of both axes used (1) points plotted correctly – tolerance ±½ square (1) straight line of best fit clearly missing point at 0.2 (1)		1 1		1 3	3 3														
		(iii)		rate proportional to concentration of I ⁻ accept first order			1	1	1														
		(iv)		rate read from graph – in the range 28-32 × 10 ⁻³ s ⁻¹ (1) time = 33 s accept range 31.25-35.7 (1)		2		2	1														
		(v)		reaction is faster at start / slows as it proceeds (1) concentrations of reactants higher at start / average rate is measured (1)			2	2	2														

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		 axes labelled (1) curve – start at origin, hump, does not touch x-axis (1)	2			2		
		(ii)		activation energy E_a is lower with a catalyst / E_a with catalyst and without catalyst marked on diagram (1) more particles / collisions exceed E_a with catalyst (1)		2		2		
				Question 12 total	2	7	4	13	6	6